Dhairya Pandya

AI/ML Engineer / Dual degree in progress
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AI/ML engineer focused end to end on machine learning: data pipelines, model training, evaluation, and deployment. Every result I report comes with the baseline it beat and the conditions under which it fails.

Open to AI/ML internships and research roles for Summer 2027.

EXPERIENCE

ML Engineer, Shipd
Freelance / Remote / 03.2026 - now

- Building ML evaluation benchmarks and data solutions for AI labs to reinforce and improve the engineering capabilities of frontier models.
- Finished in top-three positions for more than 20 challenges spanning NLP, time-series, fine-tuning, computer vision, sequence-to-sequence, from-scratch modelling, and recommendation systems.

Competitions Expert (in progress), Kaggle
Part-time / 01.2025 - now

- Finished 539th on Santa 2025, Christmas Tree Packing Challenge, a geometric optimization problem scored on normalized bounding-box area.
- Currently competing in BioHub Cell Tracking, still open.

SELECTED SYSTEMS

PathOS
Shipped / 05.2026 - now

- Offline AI pathologist for H&E histopathology screening, running the first-pass slide read on a laptop with no GPU and no internet.
- 88% yes/no accuracy on a 100-sample PathVQA validation set, against ~50% from the base Gemma 4 E2B with no pathology tuning.
- Won the Ollama Prize, Special Technology Track, Gemma 4 Good Hackathon (Kaggle x Google DeepMind), against 1,600+ teams.
- Gemma 4 / Unsloth / GRPO / LoRA / GGUF / Ollama

Papersseum
Shipped / 10.2026

- Real-time multi-agent territory game: five agents draw trails on a shared arena, and closing a loop takes everything inside it. Deterministic replays, Gym-style environment, replay viewer.
- Python / Multi-agent / Reinforcement learning

Tripwire
Prototype / 10.2026

- Research study of whether Beam, an open-weight reasoning model, cheats on impossible coding tasks, and whether its reasoning and report to the user reveal it: 616 episodes across 44 real bug fixes.
- Cheated in 66% of impossible-task episodes at low effort; a keyword rule over its reasoning found 87% of cheats, but only 6 of 33 cheats were mentioned in its note to the user.
- Python / Interpretability / Reasoning models / Evaluation

MoRPH
Prototype / 10.2026

- Research experiment: a small language model built from scratch to cut KV-cache memory, with a self-modeling training objective.
- 86x less cache than standard attention at 32k context (24 MB against about 2,013 MB) for the 170M-parameter MoRPH-S.
- PyTorch / Mamba2 / Attention / Pretraining

Swarm Immunity
Prototype / 11.2026

- Safety benchmark for bio-inspired coordination in LLM agent swarms: 483 LLM runs plus 327,600 runs of an agent-based twin.
- Leaderless evidence-sharing kept contamination at 0.06; confidence-elected leaders reached 0.67 with three of twelve agents compromised.
- Multi-agent / LLM agents / Safety / Evaluation

Udte Bhawre
Shipped / 04.2026 - 05.2026

- Cross-platform desktop AI companion living in the taskbar, with concurrent Gemini, Claude and Codex sessions

behind one MCP tool layer.
- 2nd prize, team getalife(), Hackbyte 4.0, IIITDM Jabalpur.
- Electron / Node.js / MongoDB / Gemini / MCP

RocketLandingWithRL Shipped / 09.2026 - 10.2026

- 2D rocket landing on a moving drone ship in gusty wind: environment, physics and seven RL algorithms written from scratch, trained on a laptop CPU.
- Randomized training produced a generalist that beats each level's specialist; the best mix-trained PPO lands 76% of far booster returns against a hand-written PID's 45%.
- Python / PyTorch / PPO / SAC / GRPO

NyayaSetu Shipped / 01.2026 - 03.2026

- Multi-agent legal intelligence system bridging static IPC/BNS statutes with live Supreme Court judgments, to cut hallucination in legal AI.
- 4 specialized LangGraph agents (retrieval, live-judgment verification, multimodal evidence review, synthesis), built for Convolve 4.0, a Pan-IIT hackathon.
- LangGraph / LangChain / Streamlit / Qdrant / Gemini 2.5 Flash

SKILLS AND TOOLS

Languages
- Python, Java, C++, C, SQL

ML and data
- scikit-learn, NumPy, Pandas, Matplotlib, exploratory data analysis

Serving and infra
- PostgreSQL, Hugging Face, Docker, Bash, Git

Applied AI
- Prompt engineering, retrieval, eval harnesses, quantization

Modeling
- Fine-tuning, time-series, computer vision, recommendation systems, sequence-to-sequence

Foundations
- Linear algebra, probability, statistics, optimization

VOLUNTEER

Volunteer organiser, HackByte 4.0, IIITDM Jabalpur 2026

- Helped organise a hackathon with over 6,000 registrations. Handled public relations and on-ground crowd and resource management.